INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA
(School of University of Engineering and Management, Kolkata)

End Semester Examination, November-December, 2025

Degree: **B.Tech.** Semester: **First**

Stream: **CSBS/CSE/CSE(AI & ML)/CSE(AI)/ CSE(IOT)/ CSE(IOT, CSBT)/CSE(Cyber Security)/ CSIT/CST/CSE(Data Science)/CSE(Networks)/ ECE/EEE/EE/ME/IT/Robotics & AI**

Paper Code: **ESCCS101**

Paper Name: **Introduction to AI and Data Science using Python**

Full Marks: **100** Duration: **3 Hours**

Part - A
Attempt any 10 questions out of 15 questions
Each question carries 2 Marks (2 × 10)

1. Give two examples of Generative AI tools. **2**

2. Which method is used to represent an object as a string? **2**

3. Which attribute provide us the name of the class? **2**

4. What will be the output of the following code? **2**

```
class sample:
        def __init__(self, a=0, b=0):
            self.a = a
            self.b = b
            self.display()

      def display(self):
            print(self.a, self.b)
s = sample()
```

5. What will be printed by the following code? **2**

```
x = 2
y = -1.5
```

```
z = 2 + j
product = x * y * z
print(product, type(product))
```

6. Write the output of the following program: **2**

```
a = 7
b = int(a / 2)
print(a, b)
```

7. Convert the following "while loop" into a "for loop": **2**

```
for x in range(804, 1009, 15):
        print(x * 100)
```

8. Write a Python statement to find the length of the string "Kolkata". **2**

9. What will be the output of "India".upper()? **2**

10. Which function is used to remove spaces from the beginning and end of a string? **2**

11. How do you remove an element from a set? Write the function name. **2**

12. How do you access the value of key "age" from dictionary student? **2**

13. What is the difference between read() and readline() methods? **2**

14. Which Python string function is used to replace a substring with another? For example, to replace "Java" with "Python". **2**

15. How can you check if a string is present in a larger string? **2**

Part - B

Attempt any 6 questions out of 9 questions

Each question carries 5 Marks (5 × 6)

16. Explain the working of a ChatGPT-like Generative AI model. Describe the stages: dataset collection, tokenization, neural-network training, and text generation. Explain the roles of prompt, context window, and temperature in generating responses. **5**

17. Explain the differences between local and global variables in Python. Write a Python program illustrating both types of variables. **5**

18. Create a Pandas DataFrame named StudentDat using a tuple for the following student details: **5**

No	Name	Physics	Chemistry
1001	Raghav	40	60
1002	Geetha	60	80
1003	Radha	40	60

19. Write a Python program that takes a list of numbers and prints the maximum, minimum, and average values. **5**

20. Explain how exception handling works in Python. Include the use of 'try', 'except', 'else', and 'finally' in a code example. **5**

21. Explain how Matplotlib can be used to visualize data. Include examples of line, bar, and scatter plots with short code snippets. **5**

22. Explain the concepts of encapsulation and abstraction in Python classes. Provide a simple Python class example demonstrating both concepts. **5**

23. Write a Python program to copy the contents of one file to another file. **5**

24. Write a program that fills a 3x3 matrix with values entered by the user and displays the sum of the main diagonal values. **5**

Part - C

Attempt any 5 questions out of 8 questions

Each question carries 10 Marks (10 × 5)

25. Create an one-dimensional NumPy array of 100 numbers between 0 and 1. Find out the maximum and replace that with 99. **10**

26. Create a class College with data members id, name, dept, *yearAdm* (year of admission). Take in data for 1000 members. Then input the year and print the *id*'s and *name*s of those who are now in 2nd year. **10**

27. Write a Python program that takes a set S of 100 points as input (in the form of (x,y)-coordinate pairs), and then computes and prints the largest perimeter among all triangles whose vertices are three points in the set S. **10**

28. What are 'Ideal Programming Practices with AI'? Discuss with examples how these practices ensure responsible and effective AI usage. **10**

29. Consider the Series object S that stores the contribution of each section, as shown below: **10**

D 6700
A 8000

B 5400
C 3400

i) Write code to create the series and display all its values sorted in descending order.
ii) Write code to display all its indices sorted in ascending order.

30. Write a Python program that checks whether a given list of integers can be partitioned to two lists whose elements add up to the same value. **10**

31. Given a set of (x, y) data in the form of a dictionary: {1:0.5, 2: 3.8, 3:7.9, 4:16.5, 5:27.5} . Plot y against x with x-label as 'Time' and y-label as 'Temperature'. **10**

32. Write a Python program that computes and displays the following series sum for N terms: **10**

S = 1 + 101 + 10101 + ... up to N terms.

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